📅 Project Progress — Last Two Weeks (Day 27 – Day 40)

Over the last two weeks, I focused on dataset exploration, system planning, and technical preparation for the British Sign Language recognition system. The following summarizes my progress:

🔍 Dataset Research & Exploration

- Conducted a systematic exploration of British Sign Language (BSL) datasets from multiple sources including Kaggle, GitHub repositories, and academic publications.
- Identified potential datasets such as BSL-1K, BSL Corpus Project, and BSL SignBank for comparative evaluation.
- Evaluated datasets based on dataset cardinality, class diversity, annotation quality, and modality (image vs video).
- Investigated temporal vs static gesture datasets, assessing their suitability for CNN-based classification models.
- Analyzed dataset heterogeneity, including variations in background, lighting conditions, and signer diversity.

📊 Dataset Comparison & Evaluation

- Performed comparative analysis of shortlisted datasets based on:
    - Dataset size
    - Gesture class distribution

    - Annotation format
    - Data quality
    - Real-world applicability
- Identified potential issues such as class imbalance, data sparsity, and annotation inconsistencies.
- Explored dataset augmentation feasibility to improve model generalization.

🧠 Technical Research & Model Planning

- Reviewed Convolutional Neural Network (CNN) architectures for image-based gesture recognition.
- Explored transfer learning techniques using pre-trained models such as VGG16 and ResNet.
- Investigated feature extraction pipelines and hierarchical representation learning in CNN models.
- Analyzed activation functions, including ReLU, Softmax, and Sigmoid, for multi-class classification.

👁️ Computer Vision & Preprocessing Research

- Studied image preprocessing techniques including:
    - Image normalization
    - Resizing

    - Noise reduction
    - Background segmentation
- Explored data augmentation strategies such as:
    - Rotation
    - Flipping
    - Scaling
    - Brightness adjustment
- Investigated feature standardization to improve training stability.

✋ Hand Detection & Tracking Research

- Explored hand detection frameworks including MediaPipe and OpenCV.
- Investigated hand landmark detection for gesture feature extraction.
- Evaluated real-time detection feasibility for future implementation.
- Studied pipeline integration between hand detection and CNN model.

⚙️ System Architecture Planning

- Designed a high-level system architecture, including:
    - Input capture (camera/video)
    - Hand detection module
    - Image preprocessing module

    - CNN classification model
    - Output generation (text/speech)
- Defined data flow pipeline for gesture recognition system.
- Planned modular architecture for scalability and maintainability.

⚠️ Challenges Identified

- Limited availability of British Sign Language datasets
- Dataset class imbalance
- Variation in gesture execution
- Environmental factors such as lighting and background noise
- Computational complexity for real-time gesture recognition

💡 Solutions Planned

- Dataset augmentation
- Dataset combination
- Image preprocessing
- Transfer learning
- Efficient CNN architectures

📌 Outcome of Last Two Weeks

- Dataset research completed
- Dataset comparison initiated
- Technical architecture defined
- Challenges identified
- Implementation planning started

📚 References (Harvard Style)

LeCun, Y., Bengio, Y. and Hinton, G., 2015. Deep learning. *Nature*, 521(7553), pp.436–444.

Goodfellow, I., Bengio, Y. and Courville, A., 2016. *Deep Learning*. MIT Press.

Simonyan, K. and Zisserman, A., 2015. Very deep convolutional networks for large-scale image recognition. *International Conference on Learning Representations*.

He, K., Zhang, X., Ren, S. and Sun, J., 2016. Deep residual learning for image recognition. *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*.

Zhang, F. et al., 2020. MediaPipe hands: On-device real-time hand tracking. *arXiv preprint arXiv:2006.10214*.

What went wrong?

So over the last two weeks, I've been working on setting up and building the CNN model for the British Sign Language recognition system.

I got the dataset pipeline sorted, handled the class imbalance using augmentation techniques like rotation, flipping and brightness adjustment, and started building out the

CNN architecture. I also experimented with transfer learning using VGG16 and ResNet as the base models.

However, when I actually started training the model, I ran into a significant error — essentially the model wasn't converging properly. The loss wasn't reducing the way it should, and after digging into it I realised the issue was coming from how the data was being preprocessed and fed into the model — things like inconsistent image normalisation and resizing mismatches between what MediaPipe was outputting and what the CNN expected as input.

By the time I identified the root cause, it had affected enough of the pipeline that I essentially have to rebuild and restructure parts of it from scratch to fix it properly rather than patch it.

So that's where I am right now — I've learned a lot from what went wrong, I know exactly what needs fixing, and my plan for the next two weeks is to redo that implementation cleanly with the preprocessing and data pipeline properly sorted before training again."